

Tajwar Razib

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Objective

I am a research-driven scholar with a solid background in Thermodynamics, Heat Transfer, Molecular Dynamics, Additive Manufacturing, Control System, and expertise in Machine Learning. My work combines computational and data-driven approaches to investigate energy systems and thermal processes. In an effort to broaden my scope of research, I am committed to interdisciplinary exploration and academic advancement in various scientific domains.

Education

Bachelor of Science (BSc) in Mechanical Engineering May 2026 [Expected]
Bangladesh University of Engineering and Technology (BUET)
Current CGPA: 3.70 out of 4.00 [Upto 7th Semester]

Technical Skills

- **Programming Language:** Python, C, C++
- **Software:** MATLAB, AutoCAD, SolidWorks, COMSOL, LAMMPS
- **Plot Analysis Tools:** OriginPro, Tecplot, xyExtract, Polymath Professionals

Projects

Project 1 Saylobot: Digital Data Acquisition System for Saybolt Viscometer

Here, my team automated the reading taking from the Saybolt Viscometer with the necessary sensors and Arduino program. I was the CAD designer for the project.

Project 2 Shell and Tube heat exchanger with inclined baffles

My team and I built a Shell and Tube heat exchanger with an inclined baffle. I was the Python programmer for heat exchange analysis.

Undergraduate Thesis

Machine Learning based Comparative Study of Waste Heat Recovery in Combined Organic Rankine Cycle-Gas Turbine and Conventional Steam Turbine Powerplants Applying Metaheuristic Algorithm for Working Fluid Mixture Optimization

Journal Publications

Preprints & Under Review

- **T. Razib, A. Saha.** *A Comprehensive Thermodynamic Analysis of a Bottoming Organic Rankine Cycle.* au.176124781.14639217

Manuscripts in Preparation

1. Physics Informed Disentanglement of Multimodal Data on Additive Manufacturing by Variational Auto-Encoder.

2. Comparative Analysis of PI and PID Controllers for Traction Motors in Hybrid Electrical Vehicles Using Multi-Objective Optimization via NSGA-III.
3. Analytical Analysis of LPBF using Rosenthal Thermal Equations and Application of Genetic AI.

Conference Proceedings

- Thermodynamic Analysis of a Bottoming Organic Rankine Cycle for Waste Heat Recovery.
BSME 2024
- Application of Improved Particle Swarm Optimization (PSO) on a Gas Turbine Model.
IEOM 2025
- A Comprehensive Study on Energy, COP, and Exergy of a Coupled ORC-VCC Cogeneration System Employing Dual Working Fluids.
ICME 2025

Abstract Accepted

1. Application of Bayesian Optimization on Design and Working Parameters of an Inclined baffle Shell and Tube Heat Exchanger.
ASTFE 2026

Certificates

COMSOL

Machine Learning and Deep Learning

Leadership and Outreach

Vice President, Multiscale Mechanical Modeling and Research Network (MMMRN) June 2025–Present

Organize research-related programs and networking with alumni

Treasurer, BUET Automobile Club (BAC)

March 2025–Present

Manage financial records, budgeting, and fundraising coordination.

Director, BUET Debating Club (BUETDC)

April 2025 – Present

Lead event planning and member engagement initiatives.

Vice Chairperson, G17 UAC Bangladesh

March 2024– Present

Handle recruitment, onboarding, and team communications.

Engine and Powertrain Sub-team Member, Team Automaestro September 2023 – July 2024

Worked on the design, testing, and optimization of engine components.

References

Dr. Md Ehsan

Professor, Department of Mechanical Engineering

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Dr. Anup Saha

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